EXPERIMENT - 3

CREATING DATAFRAMES & ANALYZING PROPERTIES

AIM:

To create Pandas DataFrames using NumPy arrays, CSV/Excel files and analyze the various properties of the DataFrames for real world structured data analysis.

PREREQUISITES & REQUIREMENTS:

1. Computer with Python Installed
2. Jupytor Notebook
3. Knowledge on Python, Numpy & Pandas Libraries

Step 1: Install and Import the Libraries using commands:

- pip install numpy
- pip install pandas

In [1]:
```
# importing NumPy & Pandas
import numpy as np
import pandas as pd
```

Step 2: Create a 2-D array

In [5]:
```
# Considering a 4 weeks bike sales data in three different branches
# rows --> branches, columns --> weeks
_241UDS0092_sales = np.array([[120, 220, 130, 240],
                              [166, 250, 140, 430],
                              [177, 226, 203, 540]])
```

In [6]:
```
# sales array properties
_241UDS0092_sales.shape, _241UDS0092_sales.ndim
```

Out[6]:
```
((3, 4), 2)
```

Step 3: Converting the created 2-D array into DataFrame using Pandas

In [7]:
```
_241UDS0092_df = pd.DataFrame(sales)
_241UDS0092_df
```

Out[7]:

	0	1	2	3
0	120	220	130	240
1	166	250	140	430
2	177	226	203	540

In [8]:
```
# Creating the Index & Columns for the DataFrame # Index --> Branches, Columns --> Weeks
_241UDS0092_df_index = ['Branch_1', 'Branch_2', 'Branch_3']
_241UDS0092_df_columns = ['Week_1', 'Week_2', 'Week_3', 'Week_4']
```

In [9]:
```
# Assigning the index & columns to the dataframe
_241UDS0092_df = pd.DataFrame(data=_241UDS0092_sales, index=_241UDS0092_df_index, columns=_24
```

In [10]:
```
_241UDS0092_df
```

Out[10]:

	Week_1	Week_2	Week_3	Week_4
Branch_1	120	220	130	240
Branch_2	166	250	140	430
Branch_3	177	226	203	540

In [12]:
```
# Adding Week_5 Data (Column)
# Must match number of rows (3 branches)
_241UDS0092_df['Week_5'] = [210, 220, 230]
_241UDS0092_df
```

Out[12]:

	Week_1	Week_2	Week_3	Week_4	Week_5
Branch_1	120	220	130	240	210
Branch_2	166	250	140	430	220
Branch_3	177	226	203	540	230

In [14]:
```
_241UDS0092_branch_4 = pd.DataFrame({    'Week_1': [155],
                            'Week_2': [205],
                            'Week_3': [175],
                            'Week_4': [210],
                            'Week_5': [225] }, index=['Branch_4'])
_241UDS0092_df = pd.concat([df, _241UDS0092_branch_4])
_241UDS0092_df
```

Out[14]:

	0	1	2	3	Week_1	Week_2	Week_3	Week_4	Week_5
0	120.0	220.0	130.0	240.0	NaN	NaN	NaN	NaN	NaN
1	166.0	250.0	140.0	430.0	NaN	NaN	NaN	NaN	NaN
2	177.0	226.0	203.0	540.0	NaN	NaN	NaN	NaN	NaN
Branch_4	NaN	NaN	NaN	NaN	155.0	205.0	175.0	210.0	225.0

Step 4: Creating a DataFrame with Existing CSV/EXCEL File

In [16]:
```
# Incase, if file is not in the same folder, full path is necessary
# use read_csv method
_241UDS0092_data = pd.read_csv(r"C:\Users\hi\Downloads\diesel price exp 3.csv")
_241UDS0092_data
```

Out[16]:

	Date	State	City	Diesel_Price	Petrol_Price
0	2025-01-01	Telangana	Hyderabad	95.63	107.41
1	2025-01-01	Maharashtra	Mumbai	92.15	103.50
2	2025-01-01	Karnataka	Bengaluru	88.94	101.94
3	2025-01-01	Tamil Nadu	Chennai	94.24	100.75
4	2025-01-01	Andhra Pradesh	Vijayawada	97.15	109.25
5	2025-02-01	Telangana	Hyderabad	95.63	107.41
6	2025-02-01	Maharashtra	Mumbai	92.15	103.50
7	2025-02-01	Karnataka	Bengaluru	88.94	101.94
8	2025-02-01	Tamil Nadu	Chennai	94.24	100.75
9	2025-02-01	Andhra Pradesh	Vijayawada	97.15	109.25
10	2025-03-01	Telangana	Hyderabad	95.63	107.41
11	2025-03-01	Maharashtra	Mumbai	92.15	103.50
12	2025-03-01	Karnataka	Bengaluru	88.94	101.94
13	2025-03-01	Tamil Nadu	Chennai	94.24	100.75
14	2025-03-01	Andhra Pradesh	Vijayawada	97.15	109.25
15	2025-04-01	Telangana	Hyderabad	95.63	107.41
16	2025-04-01	Maharashtra	Mumbai	92.15	103.50
17	2025-04-01	Karnataka	Bengaluru	88.94	101.94
18	2025-04-01	Tamil Nadu	Chennai	94.24	100.75
19	2025-04-01	Andhra Pradesh	Vijayawada	97.15	109.25

Step 5: Analyze the Important Properties of the DataFrames

In [18]:
```
# To view first 5 data points
_241UDS0092_data.head()
```

Out[18]:

	Date	State	City	Diesel_Price	Petrol_Price
0	2025-01-01	Telangana	Hyderabad	95.63	107.41
1	2025-01-01	Maharashtra	Mumbai	92.15	103.50
2	2025-01-01	Karnataka	Bengaluru	88.94	101.94
3	2025-01-01	Tamil Nadu	Chennai	94.24	100.75
4	2025-01-01	Andhra Pradesh	Vijayawada	97.15	109.25

In [19]:
```
# Shape of the DataFrame
_241UDS0092_data.shape
```

Out[19]:
```
(20, 5)
```

In [20]:
```
# Index Range of the DataFrame
```

```
_241UDS0092_data.index
```

Out[20]:
```
RangeIndex(start=0, stop=20, step=1)
```

In [21]:
```
# Data Types in the DataFrame
_241UDS0092_data.dtypes
```

Out[21]:
```
Date               str
State              str
City               str
Diesel_Price     float64
Petrol_Price     float64
dtype: object
```

In [22]:
```
# Retrieve the columns (feature) names of the DataFrames
_241UDS0092_data.columns
```

Out[22]:
```
Index(['Date', 'State', 'City', 'Diesel_Price', 'Petrol_Price'], dtype='str')
```

In [23]:
```
# Information about the DataFrame
_241UDS0092_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Date          20 non-null     str
 1   State         20 non-null     str
 2   City          20 non-null     str
 3   Diesel_Price  20 non-null     float64
 4   Petrol_Price  20 non-null     float64
dtypes: float64(2), str(3)
memory usage: 932.0 bytes
```

In [24]:
```
# Collecting Descriptive statistics of numerical data in a DataFrame.
# It provides summary statistics such as count, mean, standard deviation, Min etc..
```

In [25]:
```
_241UDS0092_data.describe()
```

Out[25]:

	Diesel_Price	Petrol_Price
count	20.000000	20.000000
mean	93.622000	104.570000
std	2.936116	3.328417
min	88.940000	100.750000
25%	92.150000	101.940000
50%	94.240000	103.500000
75%	95.630000	107.410000
max	97.150000	109.250000

In [26]:
```
# Transposing Rows & Columns
_241UDS0092_data.head(10).transpose()
```

Out[26]:

	0	1	2	3	4	5	6	
Date	2025-01-01	2025-01-01	2025-01-01	2025-01-01	2025-01-01	2025-02-01	2025-02-01	2025-02-0
State	Telangana	Maharashtra	Karnataka	Tamil Nadu	Andhra Pradesh	Telangana	Maharashtra	Karnatak
City	Hyderabad	Mumbai	Bengaluru	Chennai	Vijayawada	Hyderabad	Mumbai	Bengalur
Diesel_Price	95.63	92.15	88.94	94.24	97.15	95.63	92.15	88.9
Petrol_Price	107.41	103.5	101.94	100.75	109.25	107.41	103.5	101.9

RESULT:

By using Pandas, DataFrames were successfully created from NumPy arrays and CSV files. Various DataFrame properties such as shape, column names, index, data types and statistical summaries were analyzed.

In [ ]: